

2. Provedme Gram-Schmidt ortogonalizaci na $B = (1, x, x^2)$

$$v_1 = \frac{u_1}{\|u_1\|} = \frac{1}{\sqrt{\int_1^5 1^2 dx}} = \frac{1}{\sqrt{4}} = \boxed{\frac{1}{2}}$$

$$w_2 = \left(\int_1^5 \frac{1}{2} \cdot x dx \right) \cdot \frac{1}{2} = 6 \cdot \frac{1}{2} = 3$$

$$u_2 = \frac{v_2 - w_2}{\|v_2 - w_2\|} = \frac{x-3}{\sqrt{\int_1^5 (x-3)^2 dx}} = \frac{x-3}{\sqrt{\frac{16}{3}}} = \boxed{\frac{\sqrt{3}}{4} \cdot (x-3)}$$

$$w_3 = \left(\int_1^5 \frac{1}{2} \cdot x^2 dx \right) \cdot \frac{1}{2} + \left(\int_1^5 \left(\frac{\sqrt{3}}{4} (x-3) \right) \cdot x^2 dx \right) \cdot \frac{\sqrt{3}}{4} (x-3)$$

$$= \frac{31}{3} + 6(x-3) = 6x - \frac{23}{3}$$

$$u_3 = \frac{x^2 - 6x + \frac{23}{3}}{\sqrt{\int_1^5 (x^2 - 6x + \frac{23}{3})^2 dx}} = \frac{1}{\sqrt{\frac{256}{45}}} \cdot (x^2 - 6x + \frac{23}{3})$$

Pojmenujme

$$U = (u_1, u_2, u_3)$$

$$= \boxed{\frac{3\sqrt{5}}{16} \cdot (x^2 - 6x + \frac{23}{3})}$$

Nyní nalezneme OG projekci na nalezenou ON bázi.

Využijeme Důsledku 8.55.

$$[\sin x]_U = \left(\int_1^5 \frac{1}{2} \cdot \sin x dx, \int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot \sin x dx, \int_1^5 \frac{3\sqrt{5}}{16} \cdot (x^2 - 6x + \frac{23}{3}) \cdot \sin x dx \right)$$

$$\int_1^5 \frac{1}{2} \cdot \sin x \, dx \doteq 0,1283$$

$$\int_1^5 \frac{\sqrt{3}}{4} (x-3) \cdot \sin x \, dx \doteq -1,4932$$

$$\int_1^5 \frac{3\sqrt{5}}{16} \left(x^2 - 6x + \frac{23}{3}\right) \cdot \sin x \, dx \doteq -0,1252.$$

Pozn: Symbolem $\doteq$
budeme myslet zaokrouhlenou
hodnotu, kterou vrátil
software Wolfram Alpha

Upravme:

$$\begin{aligned} OG_{\text{proj.} \sin x} &\doteq 0,1283 \cdot \frac{1}{2} - 1,4932 \cdot \frac{\sqrt{3}}{4} \cdot (x-3) \\ &\quad - 0,1252 \cdot \frac{3\sqrt{5}}{16} \cdot \left(x^2 - 6x + \frac{23}{3}\right). \end{aligned}$$

Zde jsme opět použili Wolfram Alpha, aby upravil
polynom, který budeme znát $p(x)$

$$p(x) = -0,0525x^2 - 0,3315x + 1,60131.$$

Spočítáme normu

$$\|\sin x - p(x)\| = \sqrt{\int_1^5 (\sin x - p(x))^2 \, dx} \doteq 0,31879$$